

Stacked Bar Charts

100% Stacked Bar Chart

1. What is a 100% Stacked Bar Chart?

Definition

A 100% Stacked Bar Chart shows the **percentage composition** of different categories.

Key Idea

- Every bar = 100%
- Used to compare percentages, not totals

Diagram

Class A | ██████████ 80% Happy | ███ 20% Sad |
Class B | █████ 50% Happy | █████ 50% Sad |
Class C | █████ 50% Happy | █████ 50% Sad |

Each bar = 100%

Ask Students

? If one class has more students, is it fair to compare only total numbers?

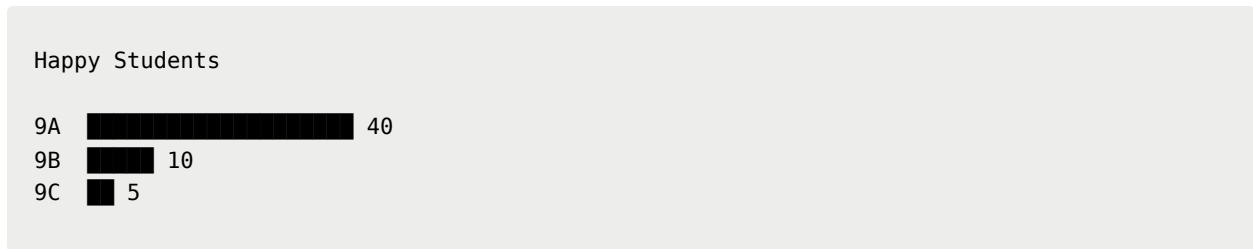
2. Why Not Use a Normal Bar Chart?

Example

Class	Students	Happy
9A	50	40

Class	Students	Happy
9B	20	10
9C	10	5

Normal Bar Chart



Observation

- 9A looks best.
- But 9A also has the most students.

Ask Students

? Is 9A happier, or does it simply have more students?

3. Convert Data into Percentages

Formula

$$\text{Percentage} = (\text{Part} \div \text{Total}) \times 100$$

Calculation

Class	Happy %
9A	80%
9B	50%
9C	50%

Diagram

$40 \div 50 \times 100 = 80\%$
 $10 \div 20 \times 100 = 50\%$
 $5 \div 10 \times 100 = 50\%$

Ask Students

? Which class is happiest now?

Answer: **9A (80%)**

4. Normal Bar Chart vs 100% Stacked Bar Chart

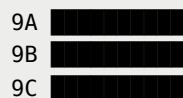
Normal Bar Chart	100% Stacked Bar Chart
Compares Numbers	Compares Percentages
Different Bar Heights	Equal Bar Heights
Shows Quantity	Shows Proportion

Diagram

Normal Chart



100% Stacked Chart



(All bars same size)

Ask Students

? Which chart is better for comparing percentages?

5. Real-Life Example

School Results

School	Students	Passed
A	1000	900
B	100	95

Using Totals

900 > 95

School A looks better.

Using Percentages

School A = 90%
School B = 95%

School B performs better.

Ask Students

? Which school would you choose based on performance?

6. When Do We Use It?

Use 100% Stacked Bar Charts For:

- ✓ Survey Results
- ✓ Exam Results
- ✓ Customer Feedback

- ✓ Sports Statistics
- ✓ Market Research

Example

Favourite Sport

Class A

Cricket 60%

Football 25%

Badminton 15%

7. Key Rule

Remember

Normal Bar Chart

=

Compare Quantities

100% Stacked Bar Chart

=

Compare Percentages

Quick Quiz

? Which chart should we use if classes have different numbers of students?

Answer: **100% Stacked Bar Chart**

8. Classroom Activity

Activity

Ask students their favourite sport:

- Cricket

- Football
- Badminton

Collect responses from:

- Class 9A
- Class 9B
- Class 9C

Steps

1. Draw a Normal Bar Chart.
2. Convert data into percentages.
3. Draw a 100% Stacked Bar Chart.
4. Compare results.

Discussion Question

? Which chart helped you understand the data better?

Final Takeaway

Normal Bar Chart
→ Compare Numbers

100% Stacked Bar Chart
→ Compare Percentages

Every Bar = 100%

This version should fit into **8–9 clean slides**, be easy for Class 9–10 students to understand, and keeps interaction every 1–2 slides so they stay engaged.